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United States Patent
Kind Code
Date of Patent
Inventor(s)

12396842
B2
August 26, 2025
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Ophthalmic lenses having an extended depth of focus for improving intermediate vision

Abstract

The present disclosure provides an ophthalmic lens (such as an IOL) that is designed to enhance depth of focus for intermediate vision performance, while maintaining distance vision. The lens may include an optic having an anterior surface and a posterior surface disposed about an optical axis. One of the surfaces (e.g., the anterior surface) may have a surface profile involving a superposition of at least three structures or profiles, including a base structure, a phase shift structure having an inner region, an outer region and a transition region, and a zonal structure having an inner power zone and an outer transition zone.

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Appl. No.: 18/407726

Filed: January 09, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240148490 A1	May. 09, 2024

Related U.S. Application Data

continuation parent-doc US 17231374 20210415 US 11903819 child-doc US 18407726
us-provisional-application US 63010792 20200416

Publication Classification

Int. Cl.: A61F2/16 (20060101)

U.S. Cl.:

CPC A61F2/1627 (20130101); A61F2/1616 (20130101); A61F2/1632 (20130101);
A61F2/164 (20150401);

Field of Classification Search

CPC: A61F (2/164)

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Background/Summary

PRIORITY CLAIM (1) This application is a continuation of U.S. patent application Ser. No. 17/231,374, filed on Apr. 15, 2021, which claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 63/010,792 titled “OPHTHALMIC LENSES HAVING AN EXTENDED DEPTH OF FOCUS FOR IMPROVING INTERMEDIATE VISION,” filed on Apr. 16, 2020, whose inventors are Myoung-Taek Choi, Sangyeol Lee, Shinwook Lee and Wangkuen Lee, which are hereby incorporated by reference in their entirety as though fully and completely set forth herein.

FIELD

(1) The present disclosure generally relates to the field of ophthalmic lenses, and more specifically

to lenses having an extended depth of focus for improving intermediate vision.

BACKGROUND

(2) Ophthalmic lenses, such as intraocular lenses, are routinely implanted in patients' eyes during cataract surgery to replace the natural crystalline lenses. The optical power of the natural crystalline lens can vary under the influence of the ciliary muscles to provide accommodation for viewing objects at different distances from the eye. Many intraocular lenses provide improved distance performance, but may lack provision for extended depth of focus for intermediate vision.

SUMMARY

(3) The present disclosure is generally directed to an ophthalmic lens (such as an IOL) that enhances depth of focus for intermediate vision performance, while maintaining distance vision.

(4) In accordance with the present disclosure, the lens includes an optic having an anterior surface and a posterior surface disposed about an optical axis, at least one of the anterior and posterior surfaces having a surface profile corresponding to a superposition of at least three profiles. The three profiles include a phase shift structure, a zonal structure, and a base structure or curvature. The phase shift structure is characterized by an inner region, an outer region, and a transition region. The inner region extends radially from the optical axis to a first boundary. The transition region is disposed between the inner region and the outer region and extends radially from the first boundary to a second boundary, wherein the transition region is adapted such that a phase of radiation incident thereon varies linearly over at least a portion of the radial extent between the first boundary and the second boundary so as to generate a phase shift between the first and second boundaries. The second boundary is disposed at a radial distance further from the optical axis than the first boundary. The outer region extends radially from the second boundary towards an outermost edge of the optic. The zonal structure includes an inner power zone having a first curvature and an outer transition zone having a second curvature. The inner power zone extends radially from the optical axis to the second boundary, and the outer transition zone extends radially to a third boundary. The third boundary is disposed at a radial distance further from the optical axis than the second boundary. The base curvature extends radially from the third boundary to the outermost edge of the optic.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a top view of an example embodiment of an ophthalmic lens, in accordance with the present disclosure;
- (2) FIG. 1B illustrates a side view of an example embodiment of an ophthalmic lens, in accordance with the present disclosure;
- (3) FIG. 2 illustrates an exploded cross-sectional view of the layers and the resultant sag of an example optic, in accordance with the present disclosure;
- (4) FIG. 3A illustrates a plot of the surface sag versus radial distance from the optical axis for a phase shift structure of an example optic, in accordance with the present disclosure;
- (5) FIG. 3B illustrates a plot of the surface sag versus radial distance from the optical axis for a zonal structure of an example optic, in accordance with the present disclosure;
- (6) FIG. 3C illustrates a plot of the composite surface sag versus radial distance from the optical axis for an example optic, in accordance with the present disclosure;
- (7) FIG. 4A illustrates a through-focus modulation transfer function (MTF) plot for a conventional monofocal IOL, in accordance with the present disclosure;
- (8) FIG. 4B illustrates a through-focus MTF plot for an optic having only a phase shift structure, in accordance with the present disclosure;
- (9) FIG. 4C illustrates a through-focus MTF plot for an optic having only a zonal power structure,

in accordance with the present disclosure;

(10) FIG. 4D illustrates a through-focus MTF plot for an example optic, in accordance with the present disclosure; and

(11) FIG. 5 illustrates a graph of visual acuity (VA) versus defocus and depicts depth of focus extensions for example optics, in accordance with the present disclosure.

(12) As understood by one of ordinary skill in the art, the drawings described below are for illustration purposes only, and are not intended to limit the scope of the present disclosure.

DETAILED DESCRIPTION

(13) Intraocular lenses (IOLs) are the most common type of lenses used with cataract surgery. Monofocal IOLs are intraocular lenses having a single point of focus for one distance, e.g., near-distance focus, mid-distance focus, or long-distance focus. Since a monofocal IOL may only be set for one distance, and because most patients and practitioners opt for long-distance focus, patients may be required to utilize additional eyewear to correct for vision at near and/or intermediate visions. Additionally, as human eyes age, depth of focus diminishes. Depth of focus (measured in diopters, D) is the total distance in front of and behind the focal point over which the image may be focused without causing a sharpness reduction beyond a certain tolerable amount. Conventional monofocal IOLs are limited in extending depth of focus.

(14) The present disclosure is generally directed to an ophthalmic lens (such as a monofocal IOL) having a surface profile that maintains distance image quality while improving intermediate vision, and further extends depth of focus. Although the following disclosure is described in conjunction with IOLs, it is to be understood that the features and elements of the present disclosure are not to be limited to any particular type of IOL and may be applied to monofocal or multifocal IOLs. Additionally, the present disclosure may further be applied to non-IOL ophthalmic lenses, such as contact lenses. Moreover, as used herein, the term “intraocular lenses” (and its abbreviation IOL) is used to describe lenses that are implanted into the interior of the eye to either replace the eye's natural lens or to otherwise augment vision regardless of whether the natural lens is removed.

(15) Reference is now made to FIGS. 1A and 1B, wherein are depicted an example embodiment of an intraocular lens **100** according to the present disclosure. FIG. 1A depicts a top view of an anterior surface **130** of the lens **100**, and FIG. 1B depicts a side view showing the anterior **130** and the posterior **140** surfaces of the lens **100**. Lens **100** may comprise a plurality of haptics **110** generally operable to position and stabilize the lens **100** within the capsular bag of a patient's eye. Lens **100** may further comprise an optic **120** having an anterior surface **130** and a posterior surface **140** that are disposed about an optical axis **105** of the lens. As shown in FIGS. 1A and 1B, optical axis **105** passes through the geometrical center of the optic **120**. One of the anterior **130** and posterior **140** surfaces may comprise either an aspheric or a spherical surface profile, and the other of the surfaces may comprise a multi-layered surface profile formed by the superposition of three profiles or structures. For purposes of illustration, the multi-layered surface profile is depicted on the anterior surface **130** of the optic **120** in FIG. 1A. However, it is to be understood that the multi-layered surface profile shown and described in conjunction with FIG. 1A may alternatively be applied on the posterior surface **140** of the optic **120**, and an aspheric or spherical surface profile may be applied on the anterior surface **130** of the optic **120**.

(16) Reference is now made to FIG. 1A in conjunction with FIG. 2. FIG. 2 illustrates an exploded cross-sectional view showing the individual layers or structures **150**, **160**, **170** which form the multi-layered surface profile of the anterior surface **130** of the optic **120**, as well as the composite or resultant sag **180** of the anterior surface **130** of the optic **120**. FIG. 1A illustrates an overhead (top) view of the composite multi-layered surface profile of the anterior surface **130** of the optic **120**, with radial boundaries **122**, **124**, **126**, **128** in FIG. 1A corresponding to the boundaries of structures **150**, **160**, **170** in FIG. 2. In FIG. 2, it is to be understood that the left and right sides of the optical axis **105** mirror each other. Therefore, the structures, boundaries, and designations shown on a given side of the optical axis **105** may apply equally to the other side of the optical axis

105.

(17) The multi-layered surface profile of the anterior surface **130** may include the superposition of at least three profiles or structures: a phase shift structure **150**; an aspheric zonal structure **160**; and an aspheric base structure **170**. The zonal structure **160** may further include the superposition of an inner power zone **162** and an outer transition zone **164**. As discussed above, each of these structures may be defined in conjunction with a plurality of radial boundaries **122**, **124**, **126**, **128** formed at increasing radial distances from the optical axis **105** of the optic **120**. The multi-layered surface profile ($Z_{\text{sub.total}}$) of the anterior surface **130** of the optic **120**—comprising the phase shift structure **150**, the zonal structure **160**, and the base structure **170**—may be defined by the following equation:

$$Z_{\text{sub.total}} = Z_{\text{sub.max}} + Z_{\text{sub.zone1}} + Z_{\text{sub.zone2}} + Z_{\text{sub.base}} \quad \text{Eq. (1),}$$

(18) wherein, $Z_{\text{sub.max}}$ represents the surface profile of the phase shift structure **150**; $Z_{\text{sub.zone1}}$ represents the surface profile of the inner power zone **162** of the zonal structure **160**; $Z_{\text{sub.zone2}}$ represents the surface profile of the outer transition zone **164** of the zonal structure **160**; and $Z_{\text{sub.base}}$ represents the surface profile of the base structure **170**.

Each of these structures **150**, **160**, **170** will be described in turn.

(19) First, the phase shift structure **150** may comprise a trapezoid phase shift (TPS) feature having an inner region **152**, a transition region **154**; and an outer region **156**. The inner region **152** may extend radially from the optical axis **105** to a first radial boundary **122**. The transition region **154** may extend radially from the first radial boundary **122** to a second radial boundary **124**, which is disposed at a radial distance further from the optical axis **105** than the first radial boundary **122**. The transition region **154** may be configured such that a phase of radiation incident thereon varies linearly over at least a portion of the radial extent between the first radial boundary **122** and the second radial boundary **124** so as to generate a phase shift between the first radial boundary **122** and the second radial boundary **124**. The outer region **156** of the phase shift structure **150** may extend radially from the second radial boundary **124** to an outermost edge **128** of the optic **120**.

(20) The trapezoid phase shift structure **150** shown in FIG. 2 may be defined by the following equation:

$$0, \quad (0 \leq r < r_1)$$

$$(21) \quad Z_{\text{aux}} = \begin{cases} \frac{r - r_1}{(r_2 - r_1)} \Delta, & (r_1 \leq r < r_2) \\ \Delta, & (r_2 \leq r) \end{cases} \quad \text{Eq. (2a)} \quad \text{wherein, } r \text{ denotes a radial distance from}$$

the optical axis **105** of the optic **120**, $r_{\text{sub.1}}$ denotes a radial distance from the optical axis **105** to the first radial boundary **122**; $r_{\text{sub.2}}$ denotes a radial distance from the optical axis **105** to the second radial boundary **124**; and Δ denotes a step height of the phase shift structure **150** relative to the inner region **152**; and wherein, Δ is defined by the following relation:

$$(22) \quad \Delta = \frac{\lambda}{(n_{\text{sub.2}} - n_{\text{sub.1}})}, \quad \text{Eq. (2b)} \quad \text{wherein, } n_{\text{sub.1}} \text{ denotes an index of refraction of material forming the optic, } n_{\text{sub.2}} \text{ denotes an index of refraction of a medium surrounding the optic, } \lambda \text{ denotes a design wavelength, and } \alpha \text{ denotes a non-integer fraction.}$$

(23) In an embodiment, the phase shift structure **150** defined by Eqs. (2a) and (2b) is characterized by a substantially linear phase shift across the transition region **154**. More specifically, the phase shift structure **150** provides a phase shift that increases linearly from the inner boundary of the transition region **154** (corresponding to the first radial boundary **122**) to the outer boundary of the transition region **154** (corresponding to the second radial boundary **124**), with the optical path difference between the inner and the outer boundaries (or between the first and second radial boundaries **122**, **124**) corresponding to a non-integer fraction of the design wavelength.

(24) In operation, the trapezoid phase shift structure **150** may produce continual focus shifts by means of a progressive wave front delay between the inner region **152** and the outer region **156**, which results in a collective depth of focus extension. The phase shift structure may produce

varying amounts of phase shift of light waves passing through the optic **120** (depending upon the region of the optic **120** the light waves pass through), and constructive interference between the light waves having varying amounts of phase shift may produce the extended depth of focus. As described below, an additional depth of focus extension may be achieved by adding the inner power zone **162** of the zonal structure **160** to the phase shift structure **150**. In this case, the phase shift structure **150** may help to mitigate the undesired pupil-dependent focal shift that may result from the enhanced depth-of-focus extension caused by the inner power zone **162**. Additionally, without the trapezoidal phase shift structure **150**, the optic **120** would essentially function as a bifocal design.

(25) In one embodiment, the radial distance, $r_{\text{sub.1}}$, of the inner region **152** of the phase shift structure **150** (from the optical axis **105** to the first radial boundary **122**) may comprise a value ranging from about 0.45 mm to 0.75 mm. The radial distance, $r_{\text{sub.2}}$, from the optical axis **105** to the second radial boundary **124** may comprise a value ranging from about 0.75 mm to 1.05 mm. Additionally, the step height, Δ , of the phase shift structure **150** may be approximately $-2.1 \mu\text{m}$.

(26) With continued reference to FIG. 1A in conjunction with FIG. 2, the zonal structure **160** may include an inner power zone **162** and an outer transition zone **164**. Inner power zone **162** may have a first curvature and may extend radially from the optical axis **105** to the second radial boundary **124** and, as shown in FIG. 2, may overlap with the inner and transition regions **152**, **154** of the phase shift structure **150**. The inner power zone **162** may comprise a refractive surface and may be disposed to enhance depth of focus for improved intermediate vision performance. Specifically, while the trapezoid phase shift structure **150** may itself have the capability to improve depth of focus and visual acuity for intermediate vision, the inner power zone **162**, having an additional higher refractive power than the base structure **170**, may yield an add power effect that, when combined with the phase shift structure **150**, may further increase depth of focus and improve intermediate vision. Moreover, the combination of the trapezoid phase shift structure **150** and the inner power zone **162** may allow the optic to flexibly perform intermediate vision control in terms of both focal distance and power. In an embodiment, the inner power zone **162** is positioned in the innermost region of the optic **120**. This positioning is particularly important to extend depth of focus up and improve visual acuity. In an embodiment, the range of add power in the inner power zone **162** may be from 0.7 D to 2.4 D. In an embodiment, the inner power zone **162** may extend depth of focus up to 2.38 D and improve visual acuity by 0.2 for pupil diameters of 2 mm to 6 mm.

(27) In an embodiment, the inner power zone **162** shown in FIG. 2 may be defined by the following equation:

$$(28) \quad Z_{\text{zone1}} = \frac{c_1 r^2}{1 + \sqrt{1 - (1 + k_1) c_1^2 r^2}} + A_4' r^4 + A_6' r^6 \quad (0 \leq r < r_2) \quad \text{Eq. (3)} \quad \text{wherein, } r \text{ denotes a radial}$$

distance from the optical axis **105** of the optic **120**, $c_{\text{sub.1}}$ denotes a base curvature of the inner power zone **162**, $k_{\text{sub.1}}$ denotes a conic constant, $r_{\text{sub.2}}$ denotes the radial distance from the optical axis **105** to the second radial boundary **124**, $A_{\text{sub.4'}}$ is a fourth order aspheric coefficient, and $A_{\text{sub.6'}}$ is a sixth order aspheric coefficient.

(29) In an embodiment, $r_{\text{sub.2}}$ may comprise a value ranging from about 0.45 mm to 0.80 mm. In an embodiment, the value of $r_{\text{sub.2}}$ in Eq. (3) for the inner power zone **162** may be substantially equivalent to the value of $r_{\text{sub.2}}$ in Eq. (2a) for the trapezoid phase shift structure **150**. In yet another embodiment, there may be some degree of variance between the value of $r_{\text{sub.2}}$ in Eq. (3) for the inner power zone **162** and the value of $r_{\text{sub.2}}$ in Eq. (2a) for the trapezoid phase shift structure **150**. Thus, in some embodiments, the position of the second radial boundary **124** may be understood as a reference point which may vary or differ with respect to the independent structures, such as the phase shift structure **150** and the zonal structure **160**, of the optic **120**. The base curvature, $c_{\text{sub.1}}$, of the inner power zone **162** may comprise a value ranging from about 19.0 to 20.2 mm.⁻¹ for a mid-power diopter value of, e.g., 21D. The conic constant, $k_{\text{sub.1}}$, may range in value from about -100 to -30. The fourth order aspheric coefficient, $A_{\text{sub.4'}}$, may

comprise a value ranging from about -6.5×10^{-4} to -1.0×10^{-4} mm.^{sup.}−4 The sixth order aspheric coefficient, A.sub.6', may comprise a value ranging from about -1.0×10^{-5} to 3.0×10^{-5} mm.^{sup.}−5 In an embodiment, the fourth and sixth order aspheric coefficients may be selected to optimize spherical aberration of the optic **120**.

(30) The zonal structure **160** may further include an outer transition zone **164** having a second curvature. As shown in the cross-sectional view of FIG. 2, the outer transition zone **164** may functionally extend radially from the second boundary **124** to the third boundary **126**, wherein the third boundary **126** is disposed at a radial distance further from the optical axis **105** than the second boundary **124**. While FIG. 2 depicts the functional boundaries of the outer transition zone **164** as extending radially from the second radial boundary **124** to the third radial boundary **126**, it is to be understood that the structure forming the outer transition zone **164** technically extends radially from the optical axis **105** to the third radial boundary **126**. However, the innermost region **164a** of the outer transition zone **164**, from the optical axis **105** to the second radial boundary **124** (depicted by the dotted lines), does not functionally contribute to the multi-layered surface profile of the anterior surface **130** of the optic **120**. In other words, only the outermost region **164b** of the outer transition zone **164** functionally contributes to the resultant sag **180** of the multi-layered surface profile of the anterior surface **130**. Outer transition zone **164** may comprise a refractive surface, and may serve to provide a smooth transition from the inner power zone **162** to the base structure **170**.

(31) In some embodiments, outer transition zone **164** may be excluded from the design of the optic **120**. In such embodiments, it is to be understood the remaining structures (e.g., phase shift structure **150**, inner power zone **162**, and/or base structure **170**) may be modified to provide appropriate transition from the inner power zone **162** to the base structure **170**. For example, the base structure **170** (described below) may be modified to functionally begin at the second radial boundary **124** (instead of the third radial boundary **126**) and may extend radially to the outermost edge **128** of the optic. By way of further example, the superposition of the outer portion **162b** of the inner power zone **162** of the zonal structure **160** and the transition zone **154** of the phase shift structure **150** may provide transition from the inner power zone **162** to the base structure **170**. It is to be understood that these and other modifications to the optic are contemplated as within the scope of the present disclosure.

(32) The outer transition zone **164** shown in FIG. 2 may be defined by the following equation:

$$(33) \quad Z_{\text{zone2}} = \frac{c_2 r^2}{1 + \sqrt{1 - (1 + k_2) c_2^2 r^2}} + A_4'' r^4 + A_6'' r^6 \quad (r_2 \leq r < r_3) \quad \text{Eq. (4)} \quad \text{wherein, } r \text{ denotes a radial}$$

distance from the optical axis **105** of the optic **120**, c.sub.2 denotes a base curvature of the outer transition zone **164**, k.sub.2 denotes a conic constant, r.sub.2 denotes the radial distance from the optical axis **105** to the second radial boundary **124**, r.sub.3 denotes the radial distance from the optical axis **105** to the third radial boundary **126**, A.sub.4'' is a fourth order aspheric coefficient, and A.sub.6'' is a sixth order aspheric coefficient.

(34) In an embodiment, the radial distance, r.sub.2, from the optical axis **105** to the second radial boundary **124** may comprise a value ranging from about 0.45 mm to 0.80 mm. In an embodiment, the value of r.sub.2 in Eq. (4) for the outer transition zone **162** may be substantially equivalent to the value of r.sub.2 in Eq. (2a) for the trapezoid phase shift structure **150**. In yet another embodiment, there may be some degree of variance between the value of r.sub.2 in Eq. (4) for the outer transition zone **162** and the value of r.sub.2 in Eq. (2a) for the trapezoid phase shift structure **150**. The radial distance, r.sub.3, from the optical axis **105** to the third radial boundary **126** may comprise a value ranging from about 0.60 mm to 1.2 mm. The base curvature, c.sub.2, of the outer transition zone **164** may comprise a value ranging from about 20.0 to 20.5 mm.^{sup.}−1 The conic constant, k.sub.2, may range in value from about −100 to −30. The fourth order aspheric coefficient, A.sub.4'', may comprise a value ranging from -6.5×10^{-4} to -1.0×10^{-4} mm.^{sup.}−3 The sixth order aspheric coefficient, A.sub.6'', may comprise a value ranging from about -1.0×10^{-5} to 3.0×10^{-5} mm.^{sup.}−5.

(35) With continued reference to FIG. 1A in conjunction with FIG. 2, the base structure **170** may comprise a third profile or structure of the multi-layered surface profile of the anterior surface **130** of the optic **120**. In some embodiments, the base structure **170** may be in the form of a base curvature. As shown in the cross-sectional view of FIG. 2, the base structure **170** may functionally extend radially from the third radial boundary **126** to the outermost edge **128** of the optic. While FIG. 2 depicts the functional boundaries of the base curvature or base structure **170** (i.e., extending radially from the third radial boundary **126** to the outermost edge **128** of the optic), it is to be understood that the base structure **170** technically extends radially from the optical axis **105** to the outermost edge **128** of the optic **120**. However, the innermost region **170a** of the base structure **170** (depicted by the dotted lines, from the optical axis **105** to the third radial boundary **126**) does not functionally contribute to the multi-layered surface profile of the anterior surface **130** of the optic **120**. In other words, only the outermost region **170b** of the base structure **170** functionally contributes to the resultant sag **180** of the multi-layered surface profile of the anterior surface **130**.

(36) In an embodiment, base structure **170** may comprise an aspheric surface profile having a base lens power, as that term is understood in the art. In an embodiment, the base structure may have an optical power ranging from -15D to +50D.

(37) The base structure **170** shown in FIG. 2 may be defined by the following equation:

(38)
$$Z_{\text{base}} = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + A_4r^4 + A_4r^4 + A_6r^6$$
 where $r \geq r_3$ Eq. (5) wherein, r denotes a radial distance from the optical axis **105** of the optic **120**, c denotes a curvature of the base structure **170**, k denotes a conic constant, $r_{\text{sub.3}}$ denotes the radial distance from the optical axis to the third boundary **126**, $A_{\text{sub.4}}$ is a fourth order aspheric coefficient, and $A_{\text{sub.6}}$ is a sixth order aspheric coefficient.

(39) In an embodiment, the radial distance, $r_{\text{sub.3}}$, from the optical axis **105** to the third radial boundary **126** may comprise a value ranging from about 0.60 mm to 1.2 mm. The base curvature, c , of the base structure **170** may comprise a value ranging from about 0.0152 mm.⁻¹ to about 0.0659 mm.⁻¹ The conic constant, k , may comprise a value ranging from about -1162 to about -19. The fourth order aspheric coefficient, $A_{\text{sub.4}}$, may comprise a value ranging from about 0.0 mm.⁻³ to about -5.3×10^{-3} mm.⁻³ The sixth order aspheric coefficient, $A_{\text{sub.6}}$, may comprise a value ranging from about 0.0 mm.⁻³ to about 1.53×10^{-4} mm.⁻⁵.

(40) With further reference to FIG. 2, and as described above, the superposition of structures **150**, **160**, **170** may result in the composite multi-layered surface profile (labeled “resultant sag” **180** in FIG. 2). The resultant sag **180** may further correspond to first, second, third, and fourth regions **182**, **184**, **186**, **188** defined by radial boundaries **122**, **124**, **126**, **128**, respectively. Each of the first, second, third, and fourth regions **182**, **184**, **186**, **188** of resultant sag **180** may result from the composite contributions of one or more of the structures **150**, **160**, **170** described above.

(41) For example, first region **182** of the resultant sag **180** may be defined as a region extending radially from the optical axis **105** to the first radial boundary **122**. In an embodiment, the first region **182** may be formed by the superposition of the inner region **152** of phase shift structure **150** and an inner portion **162a** of the inner power zone **162** of the zonal structure **160**. In an embodiment, first region **182** may not include contributions from the outer transition zone **164** of the zonal structure **160** (e.g., the innermost region **164a** of the outer transition zone **164**) or the base structure **170** (e.g., the innermost region **170a** of the base structure) because, as discussed above, the innermost region **164a** of the outer transition zone **164** and the innermost region **170a** of the base structure do not functionally contribute to the resultant sag **180** of the first region **182**. In yet another embodiment, first region **182** of the resultant sag **180** may be defined by only the inner power zone **162**, and more specifically, the inner portion **162a** of the inner power zone **162**. In other words, the inner region **152** of the phase shift structure may not functionally contribute to the resultant sag. First region **182** may comprise a first composite aspheric profile.

(42) Second region **184** of the resultant sag **180** may be defined as a region extending radially from

the first radial boundary **122** to the second radial boundary **124**. In an embodiment, the second region **184** may be formed by the superposition of the transition region **154** of phase shift structure **150** and an outer portion **162b** of the inner power zone **162** of the zonal structure **160**. In an embodiment, second region **184** may not include contributions from the outer transition zone **164** of the zonal structure **160** (e.g., the innermost region **164a** of the outer transition zone **164**) or the base structure **170** (e.g., the innermost region **170a** of the base structure) because the innermost region **164a** of the outer transition zone **164** and the innermost region **170a** of the base structure **170** do not functionally contribute to the resultant sag **180** of the second region **184**. The transition region **154** (having the single-step phase shift), when combined with the outer portion **162b** of the inner power zone **162**, may allow resultant second region **184** to serve as a transition from the first region **182** to the third region **186**. The second region **184** may comprise a second composite aspheric profile.

(43) With continued reference to FIG. 2, third region **186** of the resultant sag **180** may be defined as a region extending radially from the second radial boundary **124** to the third radial boundary **126**. In an embodiment, the third region **186** may be formed by the superposition of the outermost region **164b** of the outer transition zone **164** of the zonal structure **160** and the outer region **156** of the phase shift structure **150**. The third region **186**, which is a combination of the outermost region **164b** of the outer transition zone **164** and the outer region **156**, may serve as a transition from the second region **184** to the fourth region **188**. In an embodiment, third region **186** may not include contributions from the base structure **170** (e.g., the innermost region **170a** of the base structure) because the innermost region **170a** of the base structure does not functionally contribute to the resultant sag **180** of the third region **186**. In yet another embodiment, third region **186** may only be defined by the outer transition zone **164** of the zonal structure **160**, and more specifically, the outermost region **164b** of the outer transition zone **164**. In other words, the outer region **156** of the phase shift structure may not functionally contribute to the resultant sag. The third region **186** may comprise a third composite aspheric profile.

(44) Fourth region **188** of the resultant sag **180** may be defined as region extending radially from the third radial boundary **126** to the outermost edge **128** of the optic. In an embodiment, the fourth region **188** may be formed by the superposition of the outer region **156** of the phase shift structure **150** and the base structure **170**. In yet another embodiment, fourth region **188** may be defined by only the base structure **170**. In other words, the outer region **156** of the phase shift structure **150** may not functionally contribute to the resultant sag. The fourth region **188** may comprise a fourth composite aspheric profile.

(45) In sum, the fundamental geometry encompassed by the example optic of FIGS. 1A and 2 and defined by Equations (1) to (5) is the combination of a trapezoid phase shift structure and a zonal refractive surface having an add power effect, which together may improve intermediate vision performance and extend depth of focus, while maintaining distance vision. It is to be understood that various modifications, enhancements, and adjustments may be made to the optic **120** described herein without departing from the spirit and the scope of the disclosure.

(46) Reference is now made to FIGS. 3A-3C that show the surface profile of the anterior surface **130** of the optic **120** shown in FIGS. 1A and 2 and defined by Eqs. (1) to (5), graphically represented as plots **310**, **320**, and **330** of sag versus radial distance from the optical axis **105** of the optic **120**. Specifically, FIG. 3A depicts a sag plot **310** for the trapezoid phase shift structure (element **150** of FIG. 2) of the optic **120**. FIG. 3B depicts a unitless sag plot **320** for the zonal structure (element **160** of FIG. 2) of the optic **120**. FIG. 3C depicts a sag plot **330** of the composite multi-layered surface profile of the anterior surface **130** of the optic **120**. In all three plots of FIGS. 3A-3C, the radius is zero at optical axis **305**. As shown in the FIG. 3C, the sag curve is substantially parabolic, consistent with an aspheric lens surface. It is to be understood that FIGS. 3A-3C are shown for illustration purposes only, i.e., to show the shapes of the curves, and therefore may not be to scale and may not show the positions of the curves as a function of particular data

points and/or units of measurement.

(47) Reference is now made to FIGS. **4A-4D** that show the through-focus modulation transfer function (MTF) plots **410, 420, 430, 440** for four optics. FIG. **4A** shows an MTF plot **410** for an optic of a monofocal IOL. FIG. **4B** shows an MTF plot **420** for an optic having a trapezoid phase shift (TPS) structure (with no zonal add power structure). FIG. **4C** shows an MTF plot **430** for an optic having a zonal add power structure (with no phase shift structure). FIG. **4D** shows an MTF plot **440** of an example optic **120** based on the design shown in FIGS. **1A** and **2** and defined by Eqs. (1) to (5). These plots **410, 420, 430, 440** may be comparatively analyzed to further understand the improvements offered by the example optic design (depicted in plot **440**) of the present disclosure. For example, as shown in FIG. **4A**, an optic of a monofocal IOL lacks extended depth of focus. Next, as shown in FIG. **4B**, the optic having a trapezoid phase shift structure (with no zonal add power structure) has limited range of depth of focus and poor MTF at the defocus above the 0.5 D (myopic side) range. Third, as shown in FIG. **4C**, the optic having a zonal add power structure (with no phase shift structure) demonstrates bifocality, as evidenced by two distinct peaks at positive and negative focal shifts. This strong bifocality may cause undesired pupil-dependent focal shift and/or halos. Finally, as shown in FIG. **4D**, the example optic based on the design shown in FIGS. **1A** and **2** and defined by Eqs. (1) to (5) of the present disclosure provides extended intermediate vision without sacrificing distance vision and with no more visual disturbances than that of a monofocal IOL. As further shown in FIG. **4D**, the example optic further provides enhanced depth of focus extension and MTF performance in the intermediate vision, e.g., particularly at 1.0 D to approximately 1.5 D.

(48) Reference is now made to FIG. **5** that shows simulated monocular visual acuity (VA) plots **500** for example optics based on the design shown in FIGS. **1A** and **2** and defined by Eqs. (1) to (5). Visual acuity (VA) is the primary measure of visual function in both clinical practice and research. VA may be modeled using the intersection of the eye's modulation transfer function (MTF) with a retinal threshold function. In FIG. **5**, the solid black curve **510** represents a monofocal IOL, and the plurality of dotted curves **520, 530, 540, 550, 560** represent example optics according the present disclosure. Each of the example optics is associated with a different add-power extension, as set forth in the graph. As shown in FIG. **5**, each of the curves of the example optics **520, 530, 540, 550, 560** shows a depth of focus extension over the monofocal IOL **510**. The depth of focus extensions range from 0.42 D in example optic **520** to a maximum depth of focus extension reaching 1.38 D (shown as **570**) in example optic **560**. While there is a small tradeoff in visual acuity as the depth of focus is extended, it is to be understood that an optimal design may attempt to balance these considerations.

(49) In use, the intraocular lenses described herein are adapted to be inserted in the human eye using conventional surgical techniques modified in accordance with the present teachings. Typically, the natural crystalline lens is first removed and the IOL can be folded into a compact size for insertion through an incision or opening in the capsular bag. Following insertion, the IOL may be manipulated to assume its proper position in the capsular bag.

(50) A variety of techniques and materials may be employed to fabricate the lenses described in this disclosure. For example, the optic **120** of FIGS. **1A** and **1B** may be formed of a variety of biocompatible polymeric materials. Some suitable biocompatible materials include, without limitation, soft acrylic polymers, hydrogel, polymethymethacrylate, polysulfone, polystyrene, cellulose, acetate butyrate, or other biocompatible materials. By way of example, in an embodiment, the optic **120** may be formed of a soft acrylic polymer (cross-linked copolymer of 2-phenylethyl acrylate and 2-phenyl-ethyl methacrylate) commonly known as Acrysof®. The haptics **110** of the lenses may be formed of suitable biocompatible materials, such as those discussed above. While in some cases the optic **120** and the haptics **110** of an IOL can be fabricated as an integral unit, in other cases they can be formed separately and joined together using techniques known in the art.

(51) Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

(52) It will be appreciated that various of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. It will also be appreciated that various presently unforeseen or unanticipated alternative, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which alternatives, variations, and improvements are also intended to be encompassed by the following claims. Those having ordinary skill in the art will appreciate that various changes can be made to the above embodiments without departing from the scope of the invention.

Claims

1. An ophthalmic lens, comprising: an optic having an anterior surface and a posterior surface disposed about an optical axis, at least one of the anterior and posterior surfaces having a composite multi-layered surface profile corresponding to a superposition of at least three profiles comprising a phase shift structure, a zonal structure, and a base structure: the zonal structure having an inner power zone having a first curvature and an outer transition zone having a second curvature; and the base structure having a base curvature extending radially from the optical axis to an outermost edge of the optic.
2. The ophthalmic lens of claim 1, wherein the phase shift structure is characterized by an inner region, an outer region, and a transition region, wherein: the inner region extends radially from the optical axis to a first boundary; the transition region is disposed between the inner region and the outer region and extends radially from the first boundary to a second boundary, the second boundary disposed at a radial distance further from the optical axis than the first boundary; and the outer region extends radially from the second boundary towards the outermost edge of the optic.
3. The ophthalmic lens of claim 2, wherein the transition region is adapted such that a phase of radiation incident thereon varies linearly over at least a portion of the radial extent between the first boundary and the second boundary so as to generate a phase shift between the first and second boundaries.
4. The ophthalmic lens of claim 2, wherein the inner power zone extends radially from the optical axis to the second boundary, and the outer transition zone extends radially from the optical axis to a third boundary, the third boundary disposed at a radial distance further from the optical axis than the second boundary.
5. The ophthalmic lens of claim 4, wherein: a first region of the optic comprises a first composite aspheric profile, the first region defined from the optical axis and extending radially to the first boundary; a second region of the optic comprises a second composite aspheric profile, the second region defined from the first boundary and extending radially to the second boundary; a third region of the optic comprises a third composite aspheric profile, the third region defined from the second boundary and extending radially to the third boundary; and a fourth region of the optic comprises a fourth composite aspheric profile, the fourth region defined from the third boundary to the outermost edge of the optic.
6. The ophthalmic lens of claim 2, wherein the composite multi-layered surface profile of the at least one of the anterior and posterior surfaces is described by the following equation:
$$Z_{\text{total}} = Z_{\text{aux}} + Z_{\text{zone1}} + Z_{\text{zone2}} + Z_{\text{base}}.$$
7. The ophthalmic lens of claim 6, wherein Z_{aux} corresponds to the phase shift structure and is

$$0, \quad (0 \leq r < r_1)$$

described by the following equation: $Z_{aux} = \{ \frac{1}{(r_2 - r_1)}(r - r_1), \quad (r_1 \leq r < r_2) \}$ wherein, r denotes a radial distance from the optical axis of the optic, r_1 denotes a radial distance from the optical axis to the first boundary; r_2 denotes a radial distance from the optical axis to the second boundary; and wherein, Δ is defined by the following relation: $\Delta = \frac{1}{(n_2 - n_1)}$, wherein, n_1 denotes an index of refraction of material forming the optic, n_2 denotes an index of refraction of a medium surrounding the optic, λ denotes a design wavelength, and α denotes a non-integer fraction.

8. The ophthalmic lens of claim 7, wherein r_1 comprises a value ranging from 0.45 mm to 0.75 mm.

9. The ophthalmic lens of claim 7, wherein r_2 comprises a value ranging from 0.75 mm to 1.05 mm.

10. The ophthalmic lens of claim 6, wherein Zzone1 corresponds to the inner power zone and is described by the following equation: $Z_{zone1} = \frac{c_1 r^2}{1 + \sqrt{1 - (1 + k_1)c_1^2 r^2}} + A_4' r^4 + A_6' r^6$ ($0 \leq r < r_2$) wherein, r denotes a radial distance from the optical axis of the optic, c_1 denotes the first curvature of the inner power zone, k_1 denotes a conic constant, r_2 denotes a radial distance from the optical axis to the second boundary, A_4' is a fourth order aspheric coefficient, and A_6' is a sixth order aspheric coefficient.

11. The ophthalmic lens of claim 10, wherein r_2 comprises a value ranging from 0.45 mm to 0.80 mm.

12. The ophthalmic lens of claim 6, wherein Zzone2 corresponds to the outer transition zone and is described by the following equation: $Z_{zone2} = \frac{c_2 r^2}{1 + \sqrt{1 - (1 + k_2)c_2^2 r^2}} + A_4'' r^4 + A_6'' r^6$ ($r_2 \leq r < r_3$)

wherein, r denotes a radial distance from the optical axis of the optic, c_2 denotes the second curvature of the outer transition zone, k_2 denotes a conic constant, r_2 denotes a radial distance from the optical axis to the second boundary, r_3 denotes a radial distance from the optical axis to a third boundary, A_4'' is a fourth order aspheric coefficient, and A_6'' is a sixth order aspheric coefficient.

13. The ophthalmic lens of claim 12, wherein r_2 comprises a value ranging from 0.45 mm to 0.80 mm.

14. The ophthalmic lens of claim 12, wherein r_3 comprises a value ranging from 0.60 mm to 1.20 mm.

15. The ophthalmic lens of claim 6, wherein Zbase corresponds to the base curvature and is described by the following equation: $Z_{base} = \frac{c r^2}{1 + \sqrt{1 - (1 + k)c^2 r^2}} + A_4 r^4 + A_6 r^6$ where $r \geq r_3$ wherein, r denotes a radial distance from the optical axis of the optic, c denotes a value of the base curvature, k denotes a conic constant, r_3 denotes a radial distance from the optical axis to a third boundary, A_4 is a fourth order aspheric coefficient, and A_6 is a sixth order aspheric coefficient.

16. The ophthalmic lens of claim 15, wherein r_3 comprises a value ranging from 0.60 mm to 1.20 mm.

17. The ophthalmic lens of claim 1, wherein the inner power zone is disposed to enhance depth of focus for intermediate vision performance.

18. The ophthalmic lens of claim 17, wherein the phase shift structure is disposed to mitigate pupil-dependent focal shift effect caused by the enhanced depth of focus by the inner power zone.

19. A method of treating a vision disorder, comprising: inserting an ophthalmic lens within a capsular bag of a patient's eye, the ophthalmic lens comprising: an optic having an anterior surface and a posterior surface disposed about an optical axis, at least one of the anterior and posterior surfaces having a surface profile corresponding to: a phase shift structure characterized by an inner region, an outer region, and a transition region, wherein, the inner region extends radially from the optical axis to a first boundary; the transition region is disposed between the inner region and the

outer region and extends radially from the first boundary to a second boundary, wherein the transition region is adapted such that a phase of radiation incident thereon varies linearly over at least a portion of the radial extent between the first boundary and the second boundary so as to generate a phase shift between the first and second boundaries, the second boundary disposed at a radial distance further from the optical axis than the first boundary; and the outer region extends radially from the second boundary towards an outermost edge of the optic; a zonal structure having an inner power zone having a first curvature and an outer transition zone having a second curvature, wherein the inner power zone extends radially from the optical axis to the second boundary and the outer transition zone extends radially to a third boundary, the third boundary disposed at a radial distance further from the optical axis than the second boundary; and a base curvature extending radially from the third boundary to the outermost edge of the optic.

20. The method of claim 19, wherein: a first region of the optic comprises a first composite aspheric profile, the first region defined from the optical axis and extending radially to the first boundary; a second region of the optic comprises a second composite aspheric profile, the second region defined from the first boundary and extending radially to the second boundary; a third region of the optic comprises a third composite aspheric profile, the third region defined from the second boundary and extending radially to the third boundary; and a fourth region of the optic comprises a fourth composite aspheric profile, the fourth region defined from the third boundary to the outermost edge of the optic.
